

Programming for Economist

PS 1

Class 9

Matias B.F. Hall,
Institute of Economics,
August 2026

KØBENHAVNS UNIVERSITET



Plan for today

1. Working with problem sets
2. Intro to VSCode
3. Hints for PS1

1. Working with problem sets

- Problem sets are difficult and have many solutions
- Use code from lectures, previous exercises, friends, stack overflow etc.
 - Use what you can, modify and discard the rest
 - Important you understand why the code works as intended
- If you get stuck
 1. Read the documentation
 2. Play around with the code
 3. Look at solution
 4. Ask me/an LLM
- Highly recommended to work in groups, but write your own code

2. VSCode

1. Open VSCode and follow along on your own computer.

3. Hints for PS1

- Q1
 - To create a function, you use:
`def function_name(inputs):`
 `return output`
 - "to the power of" in python is `**`, e.g. `x**2`
- Q2
 - A string with an `f` before it is called an fstring –written `f'string'`
 - You can put references inside an f-string via `"{}"`
 - You format references by `{object:format}`
 - Use the following format for the utility-values `"6.3f"`
 - This formats floats into 6 characters wide and 3 decimals
 - `Enumerate` returns the index and value from an array

3. Hints for PS1

- Q3

- Use:

- `with open('file location and name', 'w') as f:`
 - `f.write(what you want to write)`
 - `'w'` is for write, and `f` is arbitrary but normal for `'file'`
 - Remember to specify file type here `'txt'`

- Q4

- Use:

- `with open('file location and name ', 'r') as f:`
 - `print(f.read())`

3. Hints for PS1

- Q5
 - Use code from lectures as inspiration
 - Start by defining a function computing indifference curves(x_2 as function of x_1 and utility)
 - Then create an array with x_1 's, and looping over betas and utility you compute x_2 using the function and indexing over a Boolean, to make sure points are within the constraint.
 - Useful functions
 - `np.linspace`
 - Use for-loops with `enumerate`
 - Use Jeppe's or my code for plot inspiration/cosmetics
- Q6
 - Similar to Q5 but instead of using the vectors you need to make a grid of possible x_1 's and x_2 's at the same time using `meshgrid` on the vectors.
 - Use: `ax.plot_surface`, and `ax.contour` if you don't want to compute utility yourself

3. Hints for PS1

- Q7
 - Similar to Q6 with a mesh grid and still indexing for feasible points, this time using the budget constraint.
 - Use:
 - `h = ax.scatter(inputs,c=u_grid)`
 - `fig.colorbar(h)`
 - To create the colour grid